

Revision Game Cards — 2.3 Algorithms

Print and cut along the borders. ✂ Loop cards: each player reads the bottom "Who has...?"; whoever holds the matching "I have..." reads it out, then their own clue — the chain visits every card and loops back to the start.

Loop cards (dominoes) — cut out & deal

1 / 15 I HAVE Merge sort WHO HAS... the worst-case time complexity of binary search??	2 / 15 I HAVE $O(\log n)$ WHO HAS... the data structure that is LIFO??
3 / 15 I HAVE Stack WHO HAS... the traversal that outputs a BST in ascending order??	4 / 15 I HAVE In-order traversal WHO HAS... the average-case complexity of quick sort??
5 / 15 I HAVE $O(n \log n)$ WHO HAS... the search that requires the list to be sorted??	6 / 15 I HAVE Binary search WHO HAS... the worst-case complexity of quick sort??

I HAVE

$O(n^2)$

WHO HAS...

the data structure that is FIFO??

I HAVE

Queue

WHO HAS...

the path-finding algorithm that uses a heuristic??

I HAVE

A* search

WHO HAS...

the complexity of pushing onto a stack??

I HAVE

$O(1)$

WHO HAS...

the sort that picks a pivot and partitions the list??

I HAVE

Quick sort

WHO HAS...

the best-case complexity of insertion sort??

I HAVE

$O(n)$

WHO HAS...

the traversal order Left to Right to Node??

I HAVE

Post-order traversal

WHO HAS...

the sort that repeatedly swaps adjacent elements??

I HAVE

Bubble sort

WHO HAS...

the algorithm that finds the shortest path to all nodes with no heuristic??

I HAVE

Dijkstra's algorithm

WHO HAS...

the sort that recursively splits then merges
sublists??

Taboo cards — describe the term without the banned words

Binary search

Don't say:

- sorted
- half
- middle
- log

Merge sort

Don't say:

- divide
- split
- merge
- recursive

Big O notation

Don't say:

- complexity
- time
- growth
- worst

Dijkstra's algorithm

Don't say:

- shortest
- path
- weighted
- graph

Bubble sort

Don't say:

- swap
- adjacent
- pass
- compare

Stack

Don't say:

- LIFO
- push
- pop
- last

In-order traversal

Don't say:

- left
- node
- right
- ascending

Heuristic

Don't say:

- estimate
- guess
- $h(n)$
- distance

Top Trumps — sorting & searching (play higher/lower)

Tip: for time complexity, "higher" = worse (slower). Compare the agreed stat; best card wins.

Bubble sort

Best	$O(n)$
Average	$O(n^2)$
Worst	$O(n^2)$
Space	$O(1)$
Needs Sorted	No

Insertion sort

Best	$O(n)$
Average	$O(n^2)$
Worst	$O(n^2)$
Space	$O(1)$
Needs Sorted	No

Merge sort

Best	$O(n \log n)$
Average	$O(n \log n)$
Worst	$O(n \log n)$
Space	$O(n)$
Needs Sorted	No

Quick sort

Best	$O(n \log n)$
Average	$O(n \log n)$
Worst	$O(n^2)$
Space	$O(\log n)$
Needs Sorted	No

Linear search

Best	$O(1)$
Average	$O(n)$
Worst	$O(n)$
Space	$O(1)$
Needs Sorted	No

Binary search

Best	$O(1)$
Average	$O(\log n)$
Worst	$O(\log n)$
Space	$O(1)$
Needs Sorted	Yes

Quick-fire — clue & answer (self-test / quiz-quiz-trade)

#	Clue	Answer
1	The time complexity of binary search (average/worst).	$O(\log n)$
2	The time complexity of a single loop over n items.	$O(n)$
3	The worst-case time complexity of quick sort.	$O(n^2)$
4	The guaranteed time complexity of merge sort in all cases.	$O(n \log n)$
5	The time complexity of pushing onto a stack.	$O(1)$
6	The search algorithm that needs the data sorted first.	Binary search
7	The sort that picks a pivot and partitions the list.	Quick sort
8	The traversal: Left to Node to Right.	In-order
9	The path-finding algorithm that adds a heuristic $h(n)$ to $g(n)$.	A* search
10	The data structure that is FIFO.	Queue
11	The sort that bubbles the largest value to the end each pass.	Bubble sort
12	The space complexity of merge sort.	$O(n)$